SOMAIYA VIDYAVIHAR UNIVERSITY
K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Somaiya TRUST

Natural Language Processing

N L P

Steps • Parsing • Semantic & Pragmatic Processing
Syntactic Analysis • Challenges • Strengths • Limitations • Applications

Natural Language Processing — Definition & Scope

INTRODUCTION

NLP is the sub-field of Artificial Intelligence concerned with enabling computers to **understand, interpret, manipulate and generate** human language — both written and spoken — in a meaningful and useful way.

Linguistics

Grammar, morphology, syntax, semantics, pragmatics

Computer Science

Algorithms, data structures, ML & deep learning models

Statistics & ML

Probabilistic models, neural networks, Transformers

Key Goals of NLP

- Machine Translation — translating between human languages
- Question Answering — finding precise answers naturally
- Summarisation — condensing long documents automatically
- Information Extraction — pulling structured facts from text
- Sentiment Analysis — detecting opinion and emotion in text
- Dialogue Systems — conversational agents and chatbots

The 10-Stage NLP Pipeline — Steps 1 to 5

PIPELINE · PART 1 OF 2

1 Text Acquisition

Collect raw input — web scraping, PDF extraction, speech-to-text, database reads. Handles encoding (UTF-8, Unicode) and language detection.

2 Text Cleaning

Remove HTML tags, boilerplate, special characters, duplicate whitespace. Normalise unicode, fix encoding artefacts, handle contractions.

3 Sentence Segmentation

Split cleaned text into individual sentences using punctuation rules, abbreviation lists, and ML boundary detectors (e.g. spaCy sentencizer).

4 Tokenisation

Break sentences into words or sub-word tokens. Strategies: whitespace, rule-based, BPE (GPT), WordPiece (BERT), SentencePiece (T5).

5 Morphological Analysis

Stemming (Porter, Snowball), lemmatisation (WordNet), affix stripping, compound splitting. Reduces vocabulary and normalises word forms.

Steps 6–10 →

The 10-Stage NLP Pipeline — Steps 6 to 10

PIPELINE · PART 2 OF 2

6 POS Tagging

Label each token with its grammatical role: NN (noun), VBZ (verb), JJ (adjective), DT (determiner), IN (preposition), etc.

7 Named Entity Recognition

Identify & classify named spans: PERSON, ORG, GPE, DATE, MONEY, PRODUCT. Tools: spaCy, Stanford NER, BERT-NER.

8 Syntactic Parsing

Build constituency tree (CFG/CYK/Earley) or dependency graph (arc-eager, neural). Captures grammatical structure of the sentence.

9 Semantic Analysis

Word Sense Disambiguation, Semantic Role Labelling (FrameNet, PropBank), coreference resolution, FOL / lambda-calculus representation.

10 Pragmatic Processing

Discourse coherence (RST), speech acts, Gricean maxims, presupposition, implicature, anaphora resolution, sentiment in context.

← Steps 1–5

Worked Example — Steps 1-5 (Input: raw HTML article)

Raw input: `"<p>John's dog jumped quickly over the lazy foxes.</p>"`

1 Text Acquisition	Read HTML source; detect encoding = UTF-8; detect language = English (langdetect)
2 Text Cleaning	Strip … tags → "John's dog jumped quickly over the lazy foxes."
3 Sentence Segmentation	["John's dog jumped quickly over the lazy foxes."] — 1 sentence detected
4 Tokenisation (BPE)	["John","'s","dog","jumped","quickly","over","the","lazy","foxes",".“]
5 Morphological Analysis	"jumped"→"jump" | "quickly"→"quick" | "foxes"→"fox" | "John's"→"John"

Steps 6–10 →

Worked Example — Steps 6-10 (continued)

Sentence (after cleaning): "John's dog jumped quickly over the lazy foxes."

6	POS Tagging	John/NNP 's/POS dog/NN jumped/VBD quickly/RB over/IN the/DT lazy/JJ foxes/NNS ./.
7	Named Entity Recog.	PERSON: "John" (B-PER) | All other tokens: O (outside any entity)
8	Syntactic Parse	[S [NP John's dog] [VP jumped quickly [PP over [NP the lazy foxes]]]]
9	Semantic Analysis	Agent="John's dog" | Predicate="jumped" | Manner="quickly" | Location="over the lazy foxes"
10	Pragmatic Processing	Speech act: Declarative statement. No anaphora. No implicature detected. Sentiment: Neutral.

← Steps 1–5

Core Challenges in NLP

HARD PROBLEMS

Ambiguity

Lexical: "book" (noun vs. verb) | "bank" (financial vs. river)
Syntactic: "I saw the man with the telescope" → 2 parse trees
Semantic: "He fed the dog biscuits" — who got the biscuits?
All three can co-occur in the same sentence.

World Knowledge & Common Sense

"The trophy didn't fit in the bag because it was too big."
Who is "it"? Humans: obvious. NLP models: hard.
Winograd schema challenge. Models lack grounded physical & social commonsense knowledge.

Low-Resource Languages

7,000+ languages exist; annotated data exists for ~100.
~2,500 languages are endangered.
Most SOTA models are English-centric.
Cross-lingual transfer and zero-shot learning are active research areas.

Long-Range Dependencies

"The politician who the journalist the senator attacked interviewed resigned."
Center-embedding pushes capacity of sequential models.
Transformers: quadratic attention cost for long contexts limits scalability.

Sarcasm, Irony & Figurative Language

"What a brilliant idea!" (said sarcastically)
Metaphor, idiom, hyperbole, sarcasm require pragmatic & cultural reasoning beyond surface syntax.
SemEval sarcasm tracks: models still well below human performance.

Bias & Fairness

Models trained on biased corpora reproduce & amplify societal biases.
Gender, racial and cultural stereotypes in word embeddings (word2vec, GloVe).
Mitigation: debiasing algorithms, diverse data curation, red-teaming, constitutional AI.

Strengths of Modern NLP Systems

WHAT NLP DOES WELL

Scalability

NLP models process millions of documents in minutes — impossible for human analysts. Enables automated ingestion of entire web corpora, legal archives and medical records at negligible marginal cost.

Multilingual Capability

Models like mBERT and XLM-R cover 100+ languages from a single model. Zero-shot cross-lingual transfer enables deployment in languages with no task-specific training data.

Contextual Understanding

Transformer-based models (BERT, GPT-4) capture long-range dependencies and polysemous word meaning — far surpassing early bag-of-words and n-gram approaches.

Transfer Learning

Pre-train once on large corpora; fine-tune cheaply on small domain datasets. BERT/GPT/T5 paradigm dramatically reduces task-specific data requirements — few-shot capability.

Broad Task Coverage

A single pre-trained model fine-tunes for translation, QA, summarisation, classification, NER and generation — versatile across the entire NLP task spectrum.

State-of-the-Art Performance

On SuperGLUE, SQuAD, WMT benchmarks, modern systems match or exceed human performance on many tasks — reading comprehension, MT, coreference, and more.

Weaknesses & Limitations of NLP

CURRENT GAPS

Hallucination & Factual Errors

LLMs confidently generate plausible but incorrect facts. No built-in mechanism to verify claims against external knowledge without RAG or tool-use augmentation.

Lack of True Understanding

Models excel at statistical pattern matching but lack grounded understanding. They pass benchmarks through shortcuts ('Clever Hans' effect) rather than genuine reasoning or causal inference.

High Computational Cost

Training GPT-4 scale models costs millions of dollars and consumes enormous energy. Inference latency and memory requirements limit real-time and edge/IoT deployment.

Catastrophic Forgetting

Fine-tuning a pre-trained model on new tasks degrades performance on prior tasks. Continual learning without full retraining remains an unsolved challenge (ICLR 2024).

Interpretability & Explainability

Transformer attention is not a reliable explanation of model decisions. 'Black box' nature is a barrier for regulated industries requiring audit trails (healthcare, legal, finance).

Data Hunger & Privacy

Billions of tokens needed for pre-training. Personal data in corpora poses GDPR risks. Memorisation of training examples can leak private information (Carlini et al. 2021).

NLP Applications Across Domains

REAL-WORLD DEPLOYMENTS

Virtual Assistants

Siri, Alexa, Google Assistant. Intent detection, slot filling, entity extraction and multi-turn dialogue management at billions of daily queries.

Machine Translation

Google Translate, DeepL, AWS Translate. Neural MT processes 100B+ words/day. UN adopts NMT for multilingual document production across 6 official languages.

Clinical NLP / Healthcare

ICD coding from discharge summaries, drug interaction detection, radiology report analysis. Several FDA-cleared AI products use NLP for clinical decision support.

E-Commerce & Sentiment

Amazon/Flipkart product review analysis, aspect-level sentiment, fake review detection. Personalised recommendation systems augmented with NLP-extracted features.

Education & Grading

Automated essay scoring (GRE, IELTS), intelligent tutoring (Duolingo), plagiarism detection, reading-level assessment at scale for millions of learners.

Information Retrieval

Google neural ranking, BM25+BERT hybrid retrieval, semantic search. BERT re-ranks billions of search results daily — core to modern search engine architecture.

Code Intelligence

GitHub Copilot, Code Llama, Cursor. AI-assisted code generation, documentation, bug detection and automated code review — used by 1M+ developers (2024).

Legal & Financial NLP

Contract analysis (Harvey AI), regulatory compliance, earnings call sentiment. SEC EDGAR filings processed for risk factor extraction and fraud signal detection.

Speech & Conversational AI

ASR (Whisper), speaker diarisation, call centre automation. Nuance & AWS Transcribe handle 100M+ call-centre minutes/month for enterprise customers.

01 NLP Pipeline:

10 sequential stages — Acquire → Clean → Segment → Tokenise → Morphology → POS → NER → Parse → Semantics → Pragmatics. Errors propagate downstream.

02 Parsing is Central:

Constituency (CFG/CYK) and dependency parsing form the syntactic backbone. CYK & Earley algorithms operationalise CFG parsing; PCFG adds probabilistic disambiguation.

03 Meaning is Layered:

Semantic interpretation (WSD, SRL, FOL) followed by pragmatic processing (speech acts, coreference, Gricean maxims) builds full contextual understanding.

04 Strengths & Weaknesses:

NLP excels at scale, multilingual reach, and transfer learning. Key gaps: hallucination, commonsense reasoning, interpretability, and bias remain active frontiers.

05 Broad Real-World Impact:

Virtual assistants, clinical AI, legal document analysis, code intelligence, machine translation — NLP underpins virtually every knowledge-intensive digital product.

"Language is the most massive and inclusive art we know, a mountainous and anonymous work of unconscious generations." — Edward Sapir